A 38-variable quartic equation

Strictly positive integer unknowns; 160 binary-operation abbreviations; exact degree 4.

The 38 unknowns are the following distinct names, each a positive integer:
argJ, argFuel, argAlpha, argPellF, argIndex, argTime, argX
argY, argOrdinate, argEta, argZeta, argAuxOrd, argGamma, argSquare
argScale, argMainScale, argRank, argEll, argGuard, argCube, argProduct
argFSquare, argCSquare, argFirstQ, argMainQ, argVroot, argDelta, argQSquare
argProductX, argScaledIndex, argYSquare, argQValue, argYSum, argOffset, argGammaProduct
argTimeProduct, argRankSeed, argRootSkew
Each monospaced G-name below is an abbreviation, not an additional unknown. The fixed integers β, K, μ, ω are defined after the equation.

$\texttt{G001} = \texttt{argJ} \cdot \beta$
$\texttt{G002} = 1 + \texttt{G001}$
$\texttt{G003} = \texttt{G002} \cdot \texttt{G002}$
$\texttt{G004} = \texttt{G002} \cdot \texttt{argQSquare}$
$\texttt{G005} = \texttt{argY} \cdot \texttt{argQSquare}$
$\texttt{G006} = \texttt{G002} - \texttt{argFuel}$
$\texttt{G007} = \texttt{G006} - \texttt{argAlpha}$
$\texttt{G008} = \texttt{argX} \cdot \texttt{argY}$
$\texttt{G009} = \texttt{argEta} + \texttt{argZeta}$
$\texttt{G010} = \texttt{argY} \cdot \texttt{G009}$
$\texttt{G011} = \texttt{argEta} + \texttt{argYSum}$
$\texttt{G012} = \texttt{G011} \cdot \texttt{G011}$
$\texttt{G013} = \texttt{argProductX} + \texttt{argY}$
$\texttt{G014} = 4 \cdot \texttt{G013}$
$\texttt{G015} = 3 + \texttt{G014}$
$\texttt{G016} = \texttt{G013} \cdot \texttt{G013}$
$\texttt{G017} = \texttt{G015} + \texttt{G016}$
$\texttt{G018} = \texttt{G013} \cdot \texttt{G011}$
$\texttt{G019} = \texttt{argX} + \texttt{G018}$
$\texttt{G020} = \texttt{argProductX} \cdot \texttt{argYSum}$
$\texttt{G021} = \texttt{argCube} \cdot \texttt{G001}$
$\texttt{G022} = K \cdot \texttt{G002}$
$\texttt{G023} = \texttt{argX} + \texttt{G022}$
$\texttt{G024} = \texttt{G007} \cdot \texttt{G023}$
$\texttt{G025} = \texttt{argFuel} \cdot \texttt{G002}$
$\texttt{G026} = \texttt{G024} - \texttt{G025}$
$\texttt{G027} = \texttt{argEll} + \texttt{G009}$
$\texttt{G028} = \texttt{G027} + \texttt{argOrdinate}$
$\texttt{G029} = \texttt{G009} \cdot \texttt{argGuard}$
$\texttt{G030} = \texttt{G028} - \texttt{argRootSkew}$
$\texttt{G031} = \texttt{argScale} \cdot \texttt{G030}$
$\texttt{G032} = \texttt{argY} \cdot \texttt{G001}$
$\texttt{G033} = \texttt{argGuard} \cdot \texttt{argCube}$
$\texttt{G034} = \texttt{G031} + \texttt{G032}$
$\texttt{G035} = \texttt{G034} - \texttt{G033}$
$\texttt{G036} = \texttt{argFirstQ} \cdot \texttt{argFirstQ}$
$\texttt{G037} = \texttt{argEll} \cdot \texttt{G027}$
$\texttt{G038} = 1 + \texttt{G037}$
$\texttt{G039} = \texttt{G036} - \texttt{G038}$
$\texttt{G040} = \texttt{G006} \cdot \texttt{G001}$
$\texttt{G041} = \texttt{argAlpha} + \texttt{G040}$
$\texttt{G042} = \texttt{argQSquare} - 1$
$\texttt{G043} = \texttt{argRankSeed} \cdot \texttt{G042}$
$\texttt{G044} = \omega \cdot \texttt{G002}$
$\texttt{G045} = \mu + \texttt{G044}$
$\texttt{G046} = \texttt{argJ} \cdot \texttt{G045}$
$\texttt{G047} = \texttt{G043} + \texttt{G046}$
$\texttt{G048} = \texttt{G009} - \texttt{argRank}$
$\texttt{G049} = \texttt{G048} - 1$
$\texttt{G050} = \texttt{argMainScale} \cdot \texttt{argGamma}$
$\texttt{G051} = \texttt{argProductX} \cdot \texttt{G002}$
$\texttt{G052} = \texttt{argGammaProduct} + \texttt{argProduct}$
$\texttt{G053} = \texttt{G052} - \texttt{G049}$
$\texttt{G054} = \texttt{argMainScale} \cdot \texttt{argOffset}$
$\texttt{G055} = \texttt{G015} \cdot \texttt{G053}$
$\texttt{G056} = \texttt{G054} + \texttt{G055}$
$\texttt{G057} = \texttt{argMainQ} \cdot \texttt{argMainQ}$
$\texttt{G058} = \texttt{argDelta} \cdot \texttt{argCSquare}$
$\texttt{G059} = 1 + \texttt{G058}$
$\texttt{G060} = \texttt{G057} - \texttt{G059}$
$\texttt{G061} = \texttt{argPellF} \cdot \texttt{argPellF}$
$\texttt{G062} = \texttt{argFSquare} - 1$
$\texttt{G063} = \texttt{argDelta} \cdot \texttt{G062}$
$\texttt{G064} = \texttt{argCSquare} \cdot \texttt{argIndex}$
$\texttt{G065} = \texttt{argScaledIndex} \cdot \texttt{argScaledIndex}$
$\texttt{G066} = \texttt{G065} - \texttt{argQValue}$
$\texttt{G067} = \texttt{argTime} \cdot \texttt{G011}$
$\texttt{G068} = \texttt{argTimeProduct} - \texttt{argRank}$
$\texttt{G069} = \texttt{argFSquare} \cdot \texttt{G068}$
$\texttt{G070} = \texttt{G069} - \texttt{G011}$
$\texttt{G071} = \texttt{argAuxOrd} \cdot \texttt{argAuxOrd}$
$\texttt{G072} = \texttt{argVroot} \cdot \texttt{argVroot}$
$\texttt{G073} = \texttt{argYSquare} - \texttt{argSquare}$
$\texttt{G074} = \texttt{argQValue} \cdot \texttt{G073}$
$\texttt{G075} = \texttt{argYSquare} - \texttt{G074}$
$\texttt{G076} = \texttt{G075} - 1$
$\texttt{G077} = \texttt{G039} \cdot \texttt{G039}$
$\texttt{G078} = \texttt{G060} \cdot \texttt{G060}$
$\texttt{G079} = \texttt{G076} \cdot \texttt{G076}$
$\texttt{G080} = \texttt{G066} \cdot \texttt{G066}$
$\texttt{G081} = \texttt{G077} + \texttt{G078}$
$\texttt{G082} = \texttt{G081} + \texttt{G079}$
$\texttt{G083} = \texttt{G082} + \texttt{G080}$
$\texttt{G084} = \texttt{argSquare} - \texttt{G072}$
$\texttt{G085} = \texttt{G084} \cdot \texttt{G084}$
$\texttt{G086} = \texttt{G083} + \texttt{G085}$
$\texttt{G087} = \texttt{argScale} - \texttt{G021}$
$\texttt{G088} = \texttt{G087} \cdot \texttt{G087}$
$\texttt{G089} = \texttt{G086} + \texttt{G088}$
$\texttt{G090} = \texttt{argMainScale} - \texttt{G005}$
$\texttt{G091} = \texttt{G090} \cdot \texttt{G090}$
$\texttt{G092} = \texttt{G089} + \texttt{G091}$
$\texttt{G093} = \texttt{argRank} - \texttt{G047}$
$\texttt{G094} = \texttt{G093} \cdot \texttt{G093}$
$\texttt{G095} = \texttt{G092} + \texttt{G094}$
$\texttt{G096} = \texttt{argEll} - \texttt{G020}$
$\texttt{G097} = \texttt{G096} \cdot \texttt{G096}$
$\texttt{G098} = \texttt{argGuard} - \texttt{G026}$
$\texttt{G099} = \texttt{G098} \cdot \texttt{G098}$
$\texttt{G100} = \texttt{G095} + \texttt{G097}$
$\texttt{G101} = \texttt{G100} + \texttt{G099}$
$\texttt{G102} = \texttt{argCube} - \texttt{G004}$
$\texttt{G103} = \texttt{G102} \cdot \texttt{G102}$
$\texttt{G104} = \texttt{argProduct} - \texttt{G051}$
$\texttt{G105} = \texttt{G104} \cdot \texttt{G104}$
$\texttt{G106} = \texttt{argFSquare} - \texttt{G061}$
$\texttt{G107} = \texttt{G106} \cdot \texttt{G106}$
$\texttt{G108} = \texttt{argCSquare} - \texttt{G012}$
$\texttt{G109} = \texttt{G108} \cdot \texttt{G108}$
$\texttt{G110} = \texttt{G101} + \texttt{G103}$
$\texttt{G111} = \texttt{G110} + \texttt{G105}$
$\texttt{G112} = \texttt{G111} + \texttt{G107}$

$\texttt{G113} = \texttt{G112} + \texttt{G109}$
$\texttt{G114} = \texttt{argScale} \cdot \texttt{argFirstQ}$
$\texttt{G115} = \texttt{G114} - \texttt{G035}$
$\texttt{G116} = \texttt{G115} \cdot \texttt{G115}$
$\texttt{G117} = \texttt{argMainScale} \cdot \texttt{argMainQ}$
$\texttt{G118} = \texttt{G117} - \texttt{G056}$
$\texttt{G119} = \texttt{G118} \cdot \texttt{G118}$
$\texttt{G120} = \texttt{argVroot} - \texttt{G070}$
$\texttt{G121} = \texttt{G120} \cdot \texttt{G120}$
$\texttt{G122} = \texttt{argDelta} - \texttt{G017}$
$\texttt{G123} = \texttt{G122} \cdot \texttt{G122}$
$\texttt{G124} = \texttt{G113} + \texttt{G116}$
$\texttt{G125} = \texttt{G124} + \texttt{G119}$
$\texttt{G126} = \texttt{G125} + \texttt{G121}$
$\texttt{G127} = \texttt{G126} + \texttt{G123}$
$\texttt{G128} = \texttt{argQSquare} - \texttt{G003}$
$\texttt{G129} = \texttt{G128} \cdot \texttt{G128}$
$\texttt{G130} = \texttt{argProductX} - \texttt{G008}$
$\texttt{G131} = \texttt{G130} \cdot \texttt{G130}$
$\texttt{G132} = \texttt{argScaledIndex} - \texttt{G064}$
$\texttt{G133} = \texttt{G132} \cdot \texttt{G132}$
$\texttt{G134} = \texttt{argYSquare} - \texttt{G071}$
$\texttt{G135} = \texttt{G134} \cdot \texttt{G134}$
$\texttt{G136} = \texttt{argQValue} - \texttt{G063}$
$\texttt{G137} = \texttt{G136} \cdot \texttt{G136}$
$\texttt{G138} = \texttt{G127} + \texttt{G129}$
$\texttt{G139} = \texttt{G138} + \texttt{G131}$
$\texttt{G140} = \texttt{G139} + \texttt{G133}$
$\texttt{G141} = \texttt{G140} + \texttt{G135}$
$\texttt{G142} = \texttt{G141} + \texttt{G137}$
$\texttt{G143} = \texttt{argRankSeed} - \texttt{G041}$
$\texttt{G144} = \texttt{G143} \cdot \texttt{G143}$
$\texttt{G145} = \texttt{argYSum} - \texttt{G010}$
$\texttt{G146} = \texttt{G145} \cdot \texttt{G145}$
$\texttt{G147} = \texttt{argRootSkew} - \texttt{G029}$
$\texttt{G148} = \texttt{G147} \cdot \texttt{G147}$
$\texttt{G149} = \texttt{argOffset} - \texttt{G019}$
$\texttt{G150} = \texttt{G149} \cdot \texttt{G149}$
$\texttt{G151} = \texttt{argGammaProduct} - \texttt{G050}$
$\texttt{G152} = \texttt{G151} \cdot \texttt{G151}$
$\texttt{G153} = \texttt{argTimeProduct} - \texttt{G067}$
$\texttt{G154} = \texttt{G153} \cdot \texttt{G153}$
$\texttt{G155} = \texttt{G142} + \texttt{G144}$
$\texttt{G156} = \texttt{G155} + \texttt{G146}$
$\texttt{G157} = \texttt{G156} + \texttt{G148}$
$\texttt{G158} = \texttt{G157} + \texttt{G150}$
$\texttt{G159} = \texttt{G158} + \texttt{G152}$
$\texttt{G160} = \texttt{G159} + \texttt{G154}$

The single equation is $\boxed{\texttt{G160} = 0}$.

This is an external source-mapping candidate. Its zero set has been related to an earlier 40-variable quartic; a Lean proof of the new 38-variable endpoint has not been completed.

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{47}, \quad B = \rho^{2011}, \quad \beta = B - 1,$$
$$G_m(x) = \textstyle\sum_{j=0}^{m-1} x^j, \quad \Lambda = \rho^{1005}.$$

$$\begin{aligned}\mu &= \beta - G_{951}(\rho) \\ &\quad - \rho^{960} G_{26}(\rho) - \rho^{1004}, \\ \omega &= \beta + 8G_{917}(\rho) - 4 + (2^{45} - 4)\Lambda \\ &\quad + \rho^{960} G_9(\rho) + \rho^{987} G_8(\rho) + \rho^{1004}, \\ K &= \rho^{18} + \rho^{1004} + \rho^{2020} \\ &\quad + 2\Lambda\Phi(1/\rho),\end{aligned}$$

where

$$\begin{aligned}\chi_i &= 1 + 2^{10}(1 + w_i + 2^8 n_i \\ &\quad + 2^{17} p_i + 2^{25} s_i) \quad (0 \le i \le 941),\end{aligned}$$
$$\pi(0) = 0, \quad \pi(i) = 1 + \textstyle\sum_{j=1}^{941} \left\lfloor \dfrac{2^{44} - 1 + \chi_j - \chi_i}{2^{44}} \right\rfloor \quad (1 \le i \le 941).$$

$$\begin{aligned}\Phi(x) = &\textstyle\sum_{i=0}^{941} \chi_i x^{\pi(i)} \\ &+ 2^{10} x^{969}(x^9 + 2^{17}) G_8(2x) \\ &+ 2^{18} x^{942}(x^{18} + 2^{17}) G_9(2x).\end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 942 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. For $d = 0, 1$, let

$$h(q,b) = 2q + b + 2, \quad \mathcal{S}_d = \{u_j : d_j = d,\ 0 \le j < 430\},$$
$$r_{d,q} = 205 - 199d + |\mathcal{S}_d \cap \{0, \dots, q-1\}|.$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 941$; all index ranges are fixed:

index i	tile $t_i = (w_i, n_i, p_i, s_i)$
0	$(2, 435, 3, 1)$
$j+1$ $(j = 2q + b < 430)$	$((1-d_j)r_{0,u_j}, h(q,b), d_j r_{1,u_j}, v_j)$
$433 - 2a + b$ $(a, b = 0, 1)$	$(0, 432a + b, 0, 432a + b)$
$\substack{423+2r_{d,q}+b \\ q\in\mathcal{S}_d,\ d,b=0,1}$	$(dr_{d,q}, b, (1-d)r_{d,q}, h(q,b))$
$923 + 2a + b$ $(a, b = 0, 1)$	$(0, 432a + b, 0, 438)$
927	$(2, 434, 2, 0)$
928	$(3, 436, 4, 4)$
929	$(4, 436, 4, 0)$
$496 + t$ $(t = 434, 435, 436)$	$(1, 438, 1, t)$
$933 + 3a + b$ $(a, b = 0, 1, 2)$	$(a^2, 437, b, 437)$

A closed decoder for the 430 actions. Put $D_b(A,k) = \lfloor A/b^k \rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$\mathcal{N} = \mathcal{A} \bmod 2^{636}, \quad \mathcal{T} = \lfloor \mathcal{A}/2^{636} \rfloor,$$
$$\nu_j = D_2(\mathcal{N}, j), \quad \kappa_j = (\sigma_j + 2)^{1-\nu_j} \quad (0 \le j < 430),$$
$$\sigma_j = \textstyle\sum_{k=0}^{j-1} \nu_k, \quad \Pi_j = \prod_{k=0}^{j-1} \kappa_k \quad (0 \le j \le 430),$$
$$m_q = D_2(\mathcal{N}, 430 + q), \quad \tau_q = \textstyle\sum_{k=0}^{q-1} m_k,$$
$$F_q = 14 + m_q(D_{16}(\lfloor \mathcal{T}/\Pi_{430} \rfloor, \tau_q) - 14),$$
$$u_j = (\sigma_{j+1} - (\lfloor \mathcal{T}/\Pi_j \rfloor \bmod \kappa_j)) \bmod 216,$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b+1).$$

Empty sums/products are 0/1. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$ in base 10. Its complete decimal expansion is the concatenation of these lines:

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12889260410359561746736540743515845134244260640012702033212621
95398074180784058571518488282119861023093355389406939414042892
97455773992193784795705001618777025966369137954859764532832722
75411843410809038713502633204617333246733616940964578272360901
56186216508504953649192752154519268024509952082950238865468307
22999632482769253980274091217460707829023970817244790340704449
09085581271192071656170732004937337082171141082486384269072884
72778087419663417942892520175630649792329265834490253035577533
96449689808375788457488971017043689875433162972390177936123032
89025657757489526342497438772045714099661532481123050235898973
27443609903800541199351405985754278307227777584454508422717998
33411356440933534607489268032843 7
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